

Activity 3: Cluster-Wise Student Recommendations

Machine Learning–Based Student Segmentation

1. Objective

The objective of this document is to provide targeted and actionable recommendations for each student cluster identified through machine learning (K-Means clustering). These recommendations aim to improve academic performance, mental well-being, engagement, and career readiness.

2. Cluster-Wise Recommendations

Cluster 0 – At-Risk Students

Profile Summary:

- Low academic performance
- Low attendance and assignment completion
- High stress levels
- Poor engagement and lifestyle balance

Recommended Interventions:

1. Academic Support

- Mandatory remedial classes
- One-to-one faculty mentoring
- Weekly academic progress monitoring

2. Attendance & Discipline

- Strict attendance tracking
- Early warning alerts for absenteeism

3. Mental Health Support

- Regular counseling sessions
- Stress management and mindfulness programs
- Peer support groups

4. Parental / Mentor Involvement

- Periodic progress reports
- Parent-mentor meetings for guidance

Expected Outcome:

Reduced dropout risk, improved academic stability, and better mental well-being.

Cluster 1 – High Performers

Profile Summary:

- Strong academic performance
- High engagement and attendance
- Balanced stress and lifestyle indicators
- Career-ready behavior

Recommended Interventions:

1. **Advanced Learning Opportunities**
 - Honors or advanced coursework
 - Competitive coding, research, or innovation projects
2. **Career & Placement Support**
 - Internship opportunities
 - Placement training programs
 - Industry mentorship
3. **Leadership Development**
 - Peer mentoring roles
 - Student leadership programs
 - Team lead roles in projects
4. **Recognition & Motivation**
 - Scholarships and awards
 - Public recognition of achievements

Expected Outcome:

Maximized academic excellence, leadership development, and strong career outcomes.

Cluster 2 – Career-Confused Students

Profile Summary:

- Moderate academic performance
- Inconsistent engagement
- Average stress levels
- Lack of clear career direction

Recommended Interventions:

1. **Career Guidance**
 - Career counseling sessions
 - Aptitude and interest assessments

2. Skill Development

- Skill-mapping workshops
- Exposure to multiple career paths
- Short-term certification courses

3. Engagement Improvement

- Project-based learning
- Industry exposure programs
- Alumni interaction sessions

4. Motivation & Goal Setting

- Personal development plans
- Short-term and long-term goal tracking

Expected Outcome:

Improved engagement, enhanced career clarity, and better utilization of academic potential.

3. Conclusion

Cluster-based recommendations enable personalized student support by addressing the unique needs of each group. This approach ensures efficient resource allocation, proactive risk management, and improved overall student success.